

Planetary Boundary Layer Height Using Ceilometer Data

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Abstract

The Planetary Boundary Layer (PBL) is the lowest portion of the Earth's atmosphere, where surface heating and moisture fluxes drive vigorous mixing of aerosols. Because this mixing governs particle dispersion, near-surface temperature gradients, and the triggering of convective storms, high-resolution observations of PBL height (PBLH) are critical for air-quality modeling, urban heat-island studies, and severe-weather nowcasting. Radiosondes, launched twice daily by the National Weather Service, remain the gold standard for detecting PBLH, but with a limited quantity of daily observations, they provide coarse data spatially and temporally. Ceilometers, a vertically oriented Light Detection and Ranging (LIDAR) device, provide 30-second backscatter measurements which can be used to detect features such as dust, rain, clouds, and the PBL. Automated Surface Observing Systems (ASOS) stations, located at many airports around the United States, contain CL31 ceilometers, allowing for a much denser grid of measurements. Improved measurements would both improve observational studies and real-time forecasts if properly integrated. While the PBL is defined by temperature gradients and turbulence, standard ceilometers only measure backscatter. This, paired with noisy data caused by beam attenuation, causes PBLH detection to be a challenging task. This project looked to use an array of computational techniques in order to clean and extract relevant features from the data. Chiefly, image segmentation, image labeling, automated thresholding, and wavelet transforms were used to clean, identify, and extract features from ceilometers deployed by the Community Research on Climate and Urban Science (CROCUS) integrated field laboratory. While the algorithm showed a strong bias towards the overnight residual layer (RL), it showed skill in detecting the convective PBL. Benefits to this approach include applicability for other vertically

oriented lidar devices (Doppler LIDAR, High Spectral Resolution LIDAR, etc.), robust noise handling, and strong detection of quality-interrupting features (fog, precipitation, and clouds).

2. Background

Knowledge about the depth of the PBL is important for observational and analytical studies of the atmosphere. Typical methods include instruments such as lidars, Sonic Detection and Ranging (sodar) instruments, and chiefly, radiosondes. Radiosondes measure temperature, wind, and moisture through all vertical slices in the atmosphere and, therefore, are the best fit for detecting PBLH. Despite their importance, however, there are simply not enough observations daily to properly feed numerical models. Barring special circumstances, most National Weather Service weather forecast offices only launch weather balloons twice daily, and there is a significant distance between weather forecast offices. This results in data that, while highly accurate at points, requires supplemental observations or significant interpolation to properly feed higher resolution numerical weather prediction models.

The diurnal cycle of the PBL typically consists of the convective mixing layer (ML) during the daytime and a shallow nocturnal stable layer (SL) during the overnight hours, overtopped by a residual layer (RL) from the afternoon's ML (Stull 1988). As the evolution of the PBL is complex and cyclic, continuous monitoring devices such as lidar and sodar instruments are preferred. While Doppler lidar instruments are well-suited for detecting PBLH, their costliness prevents a dense grid from being deployed. Ceilometers, by contrast, aren't quite as costly and are widely installed at ASOS stations nationwide. Aerosol backscatter is the primary measurement that ceilometers produce. When the turbulent PBL gives way to the laminar free atmosphere, aerosols are dispersed, resulting in a detectable gradient. However, the

retrieval of this feature is a non-trivial task; signal attenuation, aerosol variation, and interference from clouds and precipitation pose challenges for simple gradient detection.

With the advent of computer vision techniques, a myriad of image processing techniques can be applied to ceilometers for PBLH retrieval. Gradient (de Bruine et al. 2017) and Haar wavelet (Caicedo et al. 2020) techniques have been applied in the past for PBLH retrieval. In this project, stationary (à trous) wavelet transforms, automated thresholding, image segmentation, and image labeling are applied to Vaisala CL61 ceilometers deployed by the CROCUS integrated field laboratory to create a fully automated PBLH retrieval algorithm with robust quality control and noise handling.

3. Methods

A. Instrumentation

The instruments used to develop this algorithm were a small fleet of Vaisala CL61 ceilometers. Unlike standard ceilometers, these come with dual polarimetric capabilities, splitting each vertical beam into a vertically polarized and a cross-polarized component. These features, combined into a linear depolarization ratio, can be used to identify the general shape of an aerosol, allowing for separation between features such as dust, fog, and rain. These instruments were deployed by the CROCUS integrated field laboratory at Northeastern Illinois University (NEIU), Carruthers Center for Inner City Studies (CCICS), and Argonne's Testbed for Multiscale Observational Science (ATMOS).

The Vaisala VL61 ceilometer is an InGaAs diode laser with an average emitted power of 30mW. It is polarized into parallel and cross-parallel components and has a vertical range of 15400 meters. The reporting resolution is 4.8 meters, and it can operate in conditions above -55 degrees Celsius. The pulse length is 160 nanoseconds, and it has a beam divergence of 0.56

millirads. It features built-in cloud base height and precipitation detection; however, due to inconsistencies with the pre-packaged precipitation detection algorithm, a new algorithm was developed in this project for more accurate quality control handling.

Despite the dual polarization capabilities of these instruments, only the volume attenuated backscatter coefficient was selected for use in this algorithm to allow for more universal applications. A direction for future work is to utilize these dual polarimetric features to improve PBLH detection or detection and labeling of features (dust, smoke, clouds, etc). While a substantial network of CL61 ceilometers is not currently in place, they could potentially show stronger results for PBLH detection, as well as classification of smoke, dust, and fog events.

B. Algorithm

Beginning with a time series plot of the attenuated volume backscatter coefficient with respect to the vertical range away from the CL61 ceilometer, both a vertical and a horizontal stationary wavelet transform are applied to eight scales. Wavelet transforms, a technique not uncommon in image processing applications, deconstruct an image into different frequency components at different times or scales. Wavelets are similar in nature to a Fourier transform; however, they maintain positional data, proving valuable for applications in image processing. From the sixteen vertical and horizontal wavelets, scales that are shown to be sensitive to planetary boundary layer characteristics, but agnostic to noise, are chosen and combined into a new, smoothed image.

This smoothed and combined image is then cropped at 5000 meters. While the CL61 ceilometer is capable of measuring data above this height, it generally measures high-frequency noise or upper-level clouds. As these features are unimportant for PBLH detection, this height was chosen to avoid their interference. From here, the day is split up into 3-hourly sections for

automatic thresholding. 3-hourly periods were chosen as high-intensity features previously would interrupt the automated thresholding for the full day. This size of time slice showed to be a strong compromise for smoothness and consistency of height values without the risk of compromising full days of data.

From there, automated OTSU thresholding is applied to each time slice. OTSU thresholding is a binary threshold method that treats the image as a histogram of intensities. It automatically searches for a value that splits the histogram into roughly equal components and thresholds at that value. This method is valuable for an automated algorithm, as the aerosols that make up the PBL have different characteristics; therefore, manual thresholding is only successful on a case-by-case basis. From this thresholded image, QC handling techniques are applied, and height values are extracted.

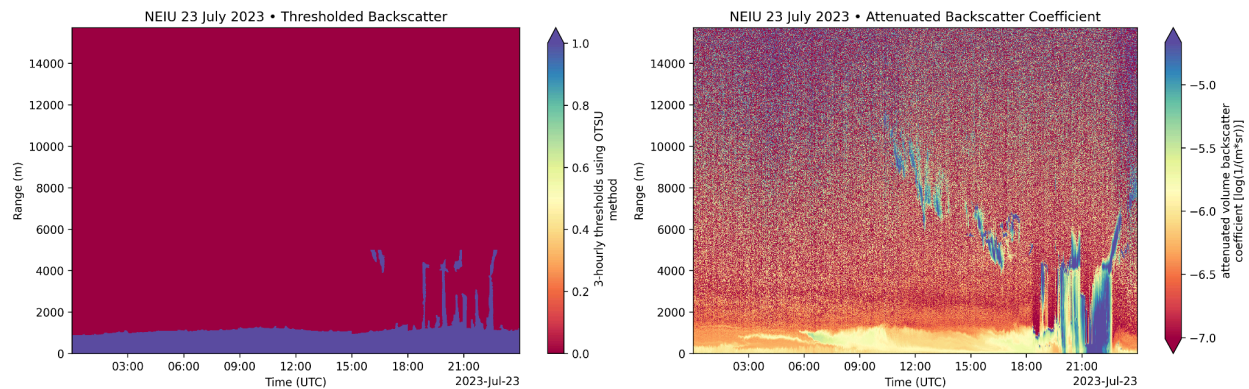


Fig 1: Thresholded Backscatter (left) and Attenuated Backscatter Coefficient (right) at NEIU on 23 July 2023

C. QC Handling

High-intensity features such as precipitation, clouds, and fog can interrupt the automated thresholding algorithm, masking subtler features. Additionally, precipitation can fully obscure the PBL. It's for these reasons that this project developed algorithms for cloud, precipitation, and fog detection. Due to the intensity of these features, detection is fairly straightforward. First, the attenuated volume backscatter coefficient is thresholded at $-5.5 \log(\text{m}^{-1}\text{sr}^{-1})$, then image

segmentation is applied to find regions of continuous high intensity features. The physical properties of these regions are then analyzed to determine their label. High intensity features which are not near the surface are labeled as clouds, high intensity features which are near the surface and extend vertically more than 200 meters are labeled as precipitation, and high intensity features which are near the surface and do not extend vertically more than 200 meters are labeled as fog. These regions are then flattened to produce flags for each time step, and timespans that are not clear air are flagged for potential QC concerns.

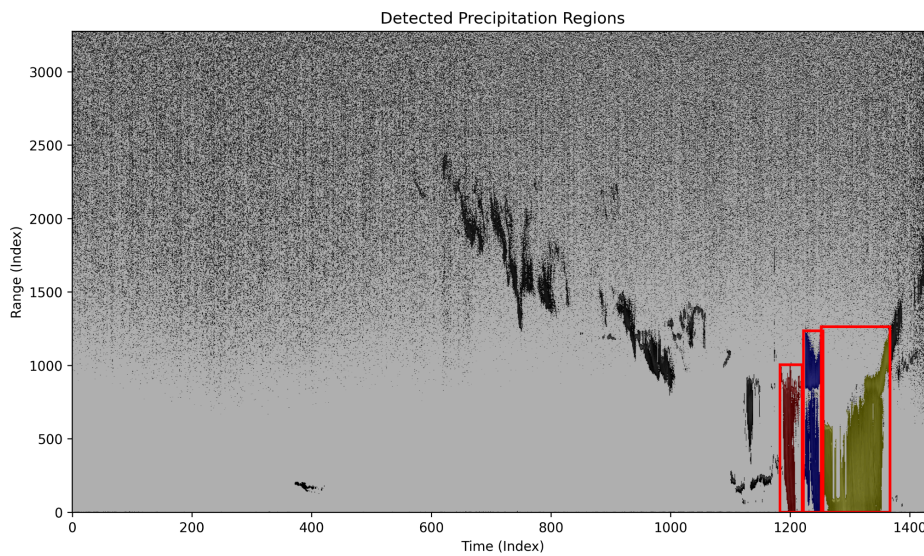


Fig 2: Detected Precipitation Regions, 23 July 2023 at NEIU

4. Results

While a validation dataset is necessary to confirm results, subjective analysis shows that the algorithm has strong promise at detecting the height of the convective mixing layer. Around a year and a half of cases were tested using data from the ceilometer located at NEIU, and barring fog and precipitation events, the convective mixing layer was well identified. What the algorithm was less successful at, however, was detecting the nocturnal stable layer. This subtle, low-to-the-ground feature was often ignored by the automated thresholding. Instead, the

algorithm tended to detect the nocturnal residual layer. Combination with an algorithm better suited for SL detection may be a future direction for this project.

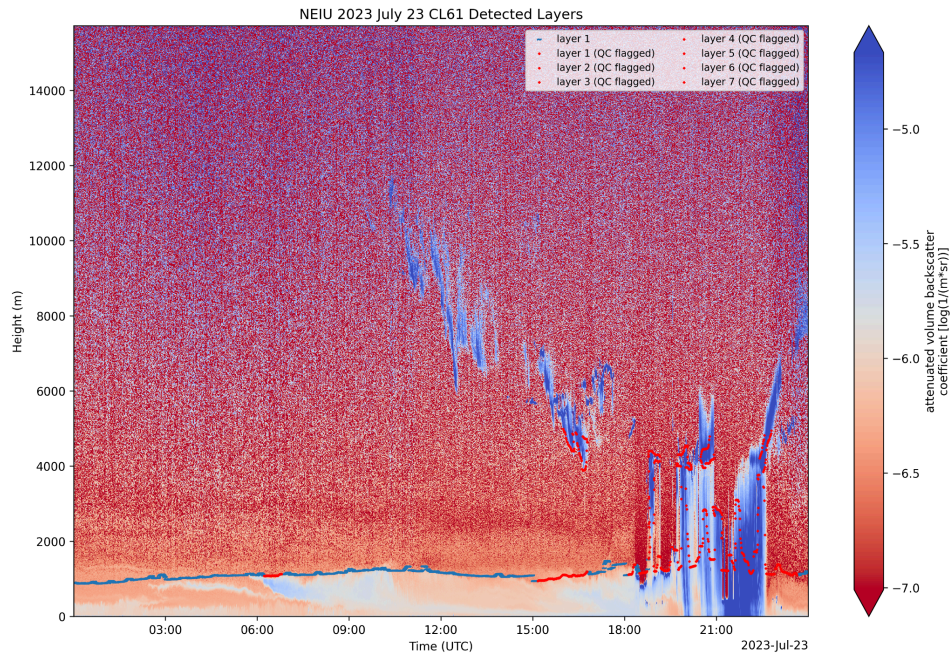


Fig 3: Detected Layers and QC flags 23 July 2023 at NEIU

5. Conclusion

While this method is computationally expensive and biased towards the residual layer, it shows strong results for diverse applications. The techniques applied in this project can be adapted for any device that outputs vertical intensity time series products. Additionally, the algorithm has very strong noise and quality control handling. Future directions for this project include developing a validation dataset utilizing Doppler lidars or radiosondes and combining these results with an algorithm that shows better results in detecting the nocturnal stable layer.

6. References

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